

→ **Entities**

◆ **Entity: Student**

- StudentID
- FirstName
- LastName
- DateOfBirth
- Age
- Email
- PhoneNumber
- GPA
- AcademicStanding
- AdvisorID

◆ **Entity: Professor**

- ProfessorID
- FirstName
- LastName
- Email
- PhoneNumber
- CollegeID
- HireDate
- Rank
- OfficeLocation

◆ **Entity: College**

- CollegeID
- CollegeName
- Building
- PhoneNumber
- ProfessorID

◆ **Entity: Program (Major)**

- ProgramID
- ProgramName
- DegreeLevel
- RequiredCredits
- Duration
- CollegeID

◆ **Entity: Course**

- CourseID
- CourseTitle
- Description
- CreditHours

- CollegeID
- PreRequisiteCourseID

◆ **Entity: Program Course**

- ProgramID
- CourseID
- RequirementType

◆ **Entity: Program Enrollment**

- ProgramEnrollmentID
- StudentID
- ProgramID
- StartDate
- EndDate
- DegreeAwarded
- EnrollmentStatus

◆ **Entity: Course Enrollment**

- CourseEnrollmentID
- StudentID
- CourseID
- ProfessorId
- AcademicYear
- AcademicTerm
- EnrollmentDate
- FinalGrade
- EnrollmentStatus

→ **Relationships**

◆ **Employs**

- College - professor
 - 1:M

◆ A college employs many professors

◆ Each professor belongs to one college

◆ **Offers**

- College - program
 - 1:M

◆ A college offers many programs

◆ Each program belongs to one college

◆ **Owns**

- College - course
 - 1:M

◆ A college manages many courses

◆ Each course belongs to one college

- ◆ Advises
 - Professor - student
 - 1:M
 - ◆ A professor advises many students
 - ◆ Each student has one advisor
- ◆ Enrolls in
 - Student - program
 - M:M
 - ◆ A student can enroll in more than one program
 - ◆ A program has many students
- ◆ Takes
 - Student - course
 - M:M
 - ◆ A student takes many courses
 - ◆ Courses has many students
- ◆ Teaches
 - Professor - course enrollment
 - 1:M
 - ◆ A professor teaches many enrollment records
 - ◆ Each record has one professor
- ◆ Includes
 - Program - course
 - M:M
 - ◆ A program requires many courses
 - ◆ A course can count towards many programs
- ◆ Is a prerequisite for
 - Course - course
 - 1:M
 - ◆ One course can be the prerequisite for many courses